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The infinitude of primes proved by the incompressibility method

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Problem: Can we use Complexity to prove theorems, though it is not computable?

Kolmogorov complexity can be used as a tool to discover mathematical proofs, by implementing a form of reasoning called the Incompressibility Method. The key idea of this method is to acknowledge the existence of incompressible strings.

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on, the exponents
r than $\log n$.

ossible exponent is
2: $n = 2^k$ and

scription, we can
 $\times \log \log n$ bits.

Infinitude of primes

1/1 point (ungraded)

What is the main argument on which Chaitin's proof of the infinitude of primes is based?

- ☐ Prime numbers that divide a number n are close to $\log n$.
- ☐ Prime numbers that divide a number n are smaller than $\sqrt{n}$.
- ☐ Prime factors can be described with $\log n$ bits, where n is the factorized number.
- ☒ Prime factorization exponents can be described with $\log q$ bits, where q is the number of bits to describe n .
- ☐ Prime factors can be described with $\log q$ bits, where q is the number of bits to describe n .
- ☐ Prime factorization is a compressing method.



Answer

Correct:

The argument is that prime factorization exponents can be described with $\log q$ bits, where q is the number of bits to describe n .

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 Answers are displayed within the problem

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